

AI-Based Badminton Stroke Technique Assessment Using Deep Learning

ITI123 Generative AI & Deep Learning Project

Project Proposal (Model Development Focus)

1 Introduction and Motivation

Badminton is a widely played sport where technical proficiency, particularly in overhead strokes, plays a critical role in player performance and injury prevention. Coaching and skill improvement in badminton traditionally rely on expert visual observation, which can be subjective, inconsistent, and difficult to scale for recreational or beginner players. Many players struggle to identify flaws in their stroke execution, such as insufficient arm extension or poor timing, resulting in slow skill development.

With recent advances in deep learning and computer vision, there is growing interest in applying AI techniques to sports analytics. However, many existing solutions focus on match-level outcomes or tactical analysis rather than stroke-level technique assessment, which is more relevant for training and skill development. This project aims to address this gap by developing an AI-based system that objectively analyses badminton stroke execution using pose-based deep learning.

The proposed approach is player-centric and applicable to both singles and doubles games, with analysis performed on a single target player per stroke. Multi-player interactions and tactical coordination are explicitly out of scope.

2 Problem Formulation

The business problem addressed in this project is the lack of accessible and objective feedback mechanisms for badminton stroke technique during training. Players require consistent and repeatable feedback to improve the execution of technically demanding overhead strokes.

This project focuses on two fundamental overhead strokes:

- Overhead Clear
- Smash

The problem is formulated as a supervised deep learning temporal sequence learning task:

- **Input:** A short video clip capturing badminton strokes (Single stroke at the beginning)
- **Intermediate Representation:** A time-ordered sequence of human body pose keypoints
- **Output:**
 - Stroke classification label (Clear or Smash)
 - Confidence score
 - Quantitative indicators describing movement characteristics such as arm extension, swing speed, and torso rotation

In doubles scenarios, the system analyses one target player per clip. Other players may appear in the frame, but the model does not perform multi-player interaction analysis.

3 Dataset and Related Research

3.1 Dataset Selection and Usage

This project leverages publicly available badminton video datasets that contain stroke-level annotations. A representative example is the ShuttleSet dataset, which provides broadcast badminton videos with labelled stroke events, including clears and smashes.

Only clips corresponding to overhead clears and smashes are used. Video data is stored locally for analysis and is not redistributed. Only derived data such as pose keypoints, extracted features, trained models, and evaluation results are included in reports or code repositories. Proper attribution to dataset authors will be provided in the final report.

4 Methodology

4.1 Pose Estimation

An open-source pose estimation model such as MoveNet or MediaPipe Pose will be used to extract 2D human body keypoints from video frames. These models are suitable for sports motion analysis and robust to variations in background and lighting commonly found in broadcast footage.

For clips containing multiple players, only the pose sequence corresponding to the target player is retained. Multi-player pose fusion and interaction modelling are out of scope.

4.2 Feature Representation

From the extracted pose keypoints, the following features are derived:

- Normalised joint coordinates
- Joint angles along the shoulder–elbow–wrist chain
- Angular velocities capturing swing dynamics
- Torso rotation indicators

These features form a temporal sequence representing individual stroke execution.

4.3 Deep Learning Architecture

The core deep learning model is a recurrent neural network designed for temporal motion analysis. Architectures explored include Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks. These architectures are computationally efficient and well-suited for short motion sequences.

5 Intended Experiments and Evaluation Plan

The experimental plan is intentionally designed to remain flexible and iterative, allowing refinement based on data quality and preliminary findings.

5.1 Experimental Approach

Planned experiments include:

- Baseline evaluation using a simple model to establish reference performance
- Primary evaluation using a recurrent deep learning model trained on pose-based temporal features
- Exploration of model and feature variants as needed, based on observed results

5.2 Evaluation Metrics

Model performance will be evaluated using standard classification metrics:

- Accuracy
- Precision, Recall, and F1-score
- Confusion matrix analysis

6 Application-Based Integration (Optional)

Application-based integration is considered a possible extension rather than a core requirement. If implemented, the demonstration will accept a video clip as input, perform pose extraction and stroke classification, and output the predicted stroke label and confidence score.

The demonstration may be implemented either as a lightweight graphical interface or as a command-line interface (CLI) tool, depending on project progress and model stability.

7 Responsible AI Considerations

Potential risks include bias arising from limited player diversity and pose ambiguity when multiple players move simultaneously. Mitigation strategies include pose normalisation, evaluation on unseen players where possible, and clear documentation of system limitations.

8 Expected Outcomes and Conclusion

The expected outcomes include a trained pose-based deep learning model for overhead stroke classification, quantitative experimental results, and a working prototype demonstrating individual stroke analysis. The system is intended as a training support tool rather than a replacement for professional coaching. Potential improvements could be considered if the initial results are satisfactory.